

Approximation Algorithms for Art Gallery Problems

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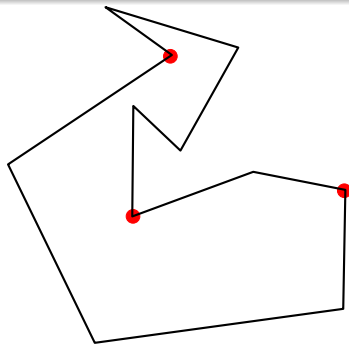
Outline

- 1 Introduction
- 2 Approximation Algorithm for Art Gallery Problem
- 3 Terrain Guarding Problem
- 3 General Terrain Guarding Problem

Introduction

Art Gallery Problem:

Given an art gallery (museum) P of a simple polygonal shape, find the minimum number of guards along with their positions inside P such that each point inside P is visible to one of the guards.



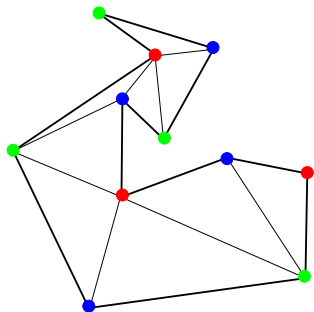
Victor Klee proposed this problem in 1976.

Introduction

Chvtal's Art Gallery Theorem

$\lfloor \frac{n}{3} \rfloor$ guards are always sufficient, and sometimes necessary to illuminate a simple polygon with n vertices.

An worst case example



Introduction

Status of the Problem

The decision version of the *art gallery problem* is **NP-complete**.

Our Objective

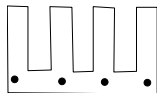
Designing an **approximation algorithm** for the *art gallery problem*.

Introduction

Variations of the Art Gallery Problem

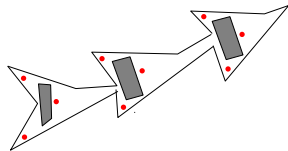
Art gallery problem for **rectilinear polygons**

Available Result: Upper bound $\lfloor \frac{n}{4} \rfloor$



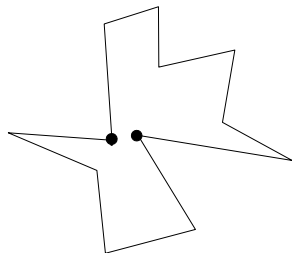
Art gallery problem with **holes**

Available Result: Upper bound $\lfloor \frac{n+2h}{3} \rfloor$
for a simple polygon with h holes and a total of n vertices.



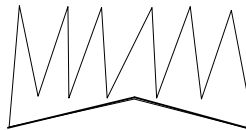
An example with 24 vertices 3 holes and 9 guards.

Vertex guard problem, where guards can only be placed at the vertices of the polygon.



Edge guard problem, where guards can patrol along the edges of the polygon.

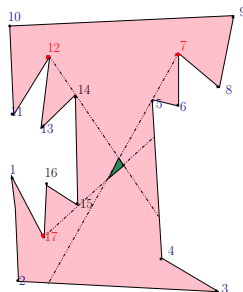
Objective: Choose minimum number of edges of the polygon where from the entire polygon is weakly visible.



Approximation Algorithm for Vertex Guard Problem

An important note

Guarding the boundary of the polygon is not enough to guard the whole polygon.



Vertices 7, 12 and 17 together can see the entire boundary of the polygon but the shaded region is not visible from any of these vertices.

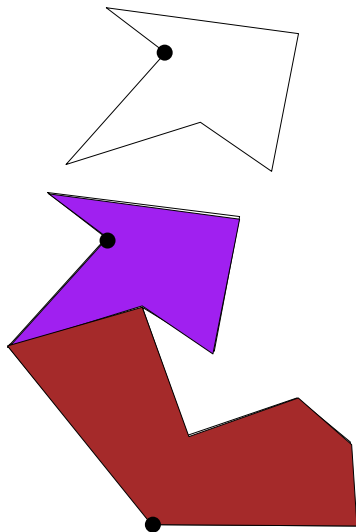
Approximation Algorithm

Definition

A simple polygon is called a fan if there exists a vertex that is visible from all points in the interior of the polygon.

Main Idea of the vertex guard problem

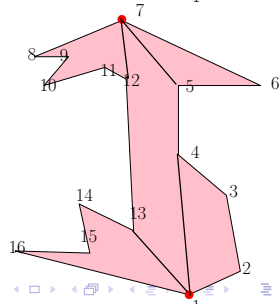
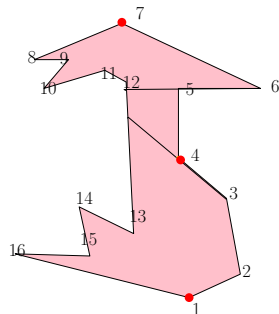
The vertex guard problem can be treated as a polygon decomposition problem in which each piece is a fan.



Approximation Algorithm

Three fans (vertices 1, 4 and 7) are necessary to cover the polygon if only edge extensions are allowed.

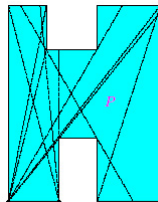
Whereas two fans (vertices 1 and 7) suffice if we allow the boundary of convex components to be bounded by segments that contains any two vertices of the polygon.



Approximation Algorithm

Arrangement $\mathcal{A}$

- Join every pair of vertices if the line segment lies inside the polygon.
- Extend them in one or both sides till it (they) hit(s) the boundary of the polygon.



Observation:

- Every cell in the arrangement $\mathcal{A}$ of these line segments inside the polygon is convex,
- Each cell in $\mathcal{A}$ is either totally visible or totally invisible from a vertex of the polygon.
- Each cell in $\mathcal{A}$ must lie in at least one of the fans corresponding to the vertices of the polygon.

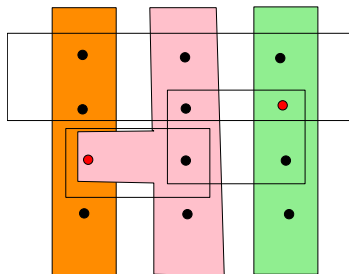
Set Cover Problem

Set Cover Problem

Given:

- $\mathcal{C} = \{c_1, c_2, \dots, c_n\}$ a set of elements, and
- $\mathcal{S} = \{S_1, S_2, \dots, S_k\}$ be a set of subsets of $\mathcal{C}$

Objective: Choose the minimum number of members in $\mathcal{S}$ that covers all the elements in $\mathcal{C}$.



Formulation as a Set Cover Problem

Formulation of the Art Gallery Problem

- $\mathcal{C}$ = The convex cells of the arrangement.
- $\mathcal{S}$ = Set of vertices of the polygon.

Each member $S_i \in \mathcal{S}$ consists of those convex cells that lie in the fan of the corresponding vertex of P .

Choose the minimum number of vertices of P whose fans cover all the cells of the arrangement $\mathcal{C}$.

Results

Approximation Factor: $\log n$

using the existing $O(\log n)$ factor approximation algorithm for the set cover problem.

Time Complexity: $O(n^5)$

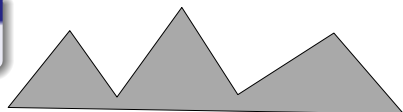
- Number of cells on the arrangement $\mathcal{A} = O(n^4)$ (worst case)
- Computing each subset S_i needs $O(n)$ time.

Terrain Guarding Problem

Terrain

An x -monotone polychain.

A vertical line intersects the polychain in exactly one point.

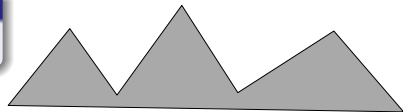


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The Problem

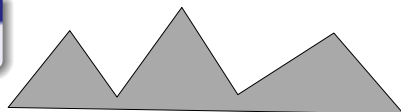
Find minimum number of points on the boundary of the terrain that can see all the vertices of the terrain.

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The Problem

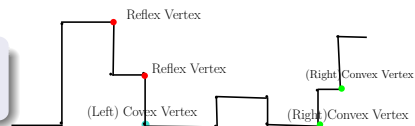
Find minimum number of points on the boundary of the terrain that can see all the vertices of the terrain.

Status of the Problem - NP-hard

See J. King and E. Krohn, *Terrain Guarding is NP-Hard*, SODA-2009.

Rectilinear Terrain

A simpler version of the problem is the **rectilinear terrain**,



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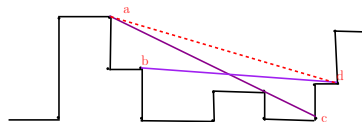


Definitions

- A terrain $\mathcal{T}$ is **rectilinear terrain** if each edge $e \in E(\mathcal{T})$ is either horizontal or vertical.
- A vertex v_i of $\mathcal{T}$ is **convex** if the angle formed by the edges e_{i-1} and e_i above $\mathcal{T}$ is 90° .
- A vertex v_i of $\mathcal{T}$ is **reflex** if the angle formed by the edges e_{i-1} and e_i above $\mathcal{T}$ is 270° .
- A convex vertex v_i is called **left convex** if the edge e_{i-1} is vertical.
- A convex vertex v_i is called **right convex** if the edge e_i is vertical.

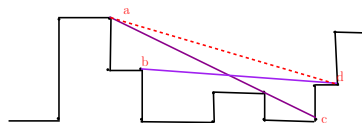
Properties of the Rectilinear Terrain

Let a , b , c and d be four points on a terrain $\mathcal{T}$, such that $a_x < b_x < c_x < d_x$. Now, if a can see c and b can see d , then a can see d .

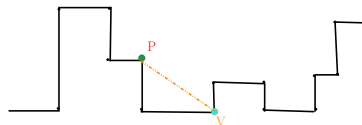


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Let v be a right convex vertex on a rectilinear terrain $\mathcal{T}$, and p be a point on $\mathcal{T}$. If p can see v , then $p_x \leq v_x$.



Properties of the Rectilinear Terrain

- If a set G of points on a rectilinear terrain $\mathcal{T}$ guards a subset $V' \subseteq V_\ell \cup V_r$, where V_ℓ (resp. V_r) is a subset of left-convex (resp. right convex) vertices of $\mathcal{T}$, then there exists a $\hat{G} \subseteq V$ of reflex vertices, such that $\hat{G}$ guards V' , and $|\hat{G}| \leq |G|$.

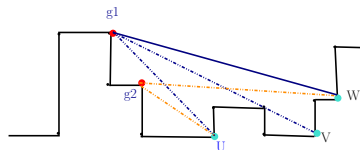
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- If $G \subseteq V$ guards all the convex vertices of a rectilinear terrain $\mathcal{T}$, then G guards all the vertices of $\mathcal{T}$, and all the vertical edges of $\mathcal{T}$.

Properties of the Rectilinear Terrain

Consider a situation where

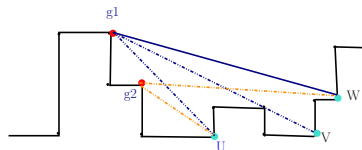
- u , v and w are three **right-convex** vertices such that $u_x < v_x < w_x$.
- There exists two vertices g_1 and g_2 of the terrain $\mathcal{T}$ such that g_1 can see both u and v and g_2 can see both u and w .



Properties of the Rectilinear Terrain

Consider a situation where

- u , v and w are three **right-convex** vertices such that $u_x < v_x < w_x$.
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then

$\exists$ a vertex g that can see all three vertices u , v and w .

Moreover, g is one among g_1 and g_2
(the one that precedes the other in the sequence of vertices defining $\mathcal{T}$).

Splitting the Problem

P1: Covering the right-convex vertices, and

P2: Covering the left-convex vertices.

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Our Strategy

- Compute optimum solution of both the problems **P1** and **P2**.

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Our Strategy

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Result

A $2 \times OPT$ solution.

Solving Problem **P1**

Construct a graph G^r

- Its vertices are the the right-convex vertices V_r of $\mathcal{T}$.
- Put an edge between a pair of vertices $u, v \in V_r$ if and only if there exists a vertex g of the terrain $\mathcal{T}$ that can see both u and v .

Solving Problem **P1**

Construct a graph G^r

- Its vertices are the the right-convex vertices V_r of $\mathcal{T}$.
- Put an edge between a pair of vertices $u, v \in V_r$ if and only if there exists a vertex g of the terrain $\mathcal{T}$ that can see both u and v .

Results

- G^r is a chordal graph.
- Size of the minimum clique cover of G^r is the minimum number of vertex guards for covering all the right-convex vertices in $\mathcal{T}$.
- Minimum clique cover of G^r can be computed in polynomial time.

General Terrain Guarding Problem

Simpler Variation: Left Guarding

We are given two sets of points N and G . Each point in N is to be guarded using only guards from G which are strictly to its left.

Linear programming formulation: LP1

minimize $\sum_{g \in G} x_g$
 subject to $\sum_{g \in S_L(p)} x_g \geq 1 \quad \forall p \in N$
 (* $S_L(p)$ is a subset of G where from $p \in N$ can be guarded *)
 $x_g \in \{0, 1\} \quad \forall g \in G$

Important Note

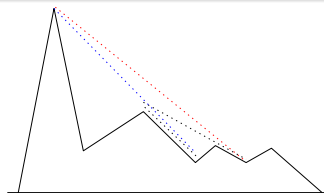
In the definition of left-guarding or right-guarding the guard does not see the point on which it lies

Optimal Solution for Left Guarding

Important Property

Any left-visibility matrix is totally balanced. In other words, there is no submatrix of the form $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

	g_1	g_2	$\dots$	g_n
n_1				
n_2				
$\vdots$				
n_m				

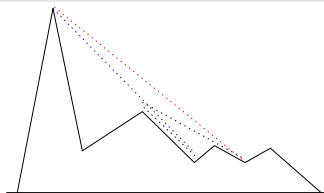


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	g_1	g_2	$\dots$	g_n
n_1				
n_2				
$\vdots$				
n_m				



Result

^a There is an efficient combinatorial algorithm for finding an optimal integral solution to (LP1)

^aA. Kolen. *Location problems on trees and in the rectilinear plane*. PhD thesis, Mathematisch Centrum, Amsterdam, 1982.

Optimal Algorithm for Left Guarding

left-guarding (T, N, G)

- $A \leftarrow \Phi$
- for $p \in N$ do (* process members in N from left to right *)
 - if p is not yet seen by the existing members in A then
 - $A \leftarrow A \cup \{L(p)\}$ (* append (left-most member) $L(p)$ to A *)
- Return A

Result

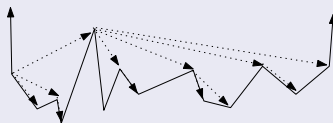
The algorithm can be implemented in $O(|N| \log |G|)$ time using a procedure similar to Graham's scan

One Sided Guarding

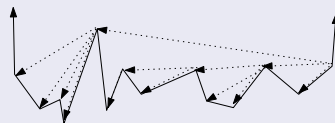
Problem Statement

We are given a set of points N and two sets of guards G_L and G_R , where each guard in G_L (respectively, G_R) can only guard points of N strictly from its right (respectively, strictly from its left).

Right-Guarding



Left-Guarding



One Sided Guarding

Integer programming formulation:LP2

$$\text{minimize } \sum_{g \in G_L} x_{g,L} + \sum_{g \in G_R} x_{g,R}$$

$$\text{subject to } \sum_{g \in S_L(p) \cap G_L} x_{g,L} + \sum_{g \in S_R(p) \cap G_R} x_{g,R} \geq 1, \forall p \in N$$

$$x_{g,L} \in \{0, 1\}, \forall g \in G_L$$

$$x_{g,R} \in \{0, 1\}, \forall g \in G_R$$

2-factor approximation Algorithm for One Sided Guarding

Step1 Find an optimal fractional solution x^* to (LP2)

Step2 Guided by x^* , we divide the points into two sets:

$$N_L = \{p \in N \mid \sum_{g \in S_L(p) \cap G_L} X_{g,L}^* \geq \frac{1}{2}\}, \text{ and}$$

$$N_R = \{p \in N \mid \sum_{g \in S_R(p) \cap G_R} X_{g,R}^* \geq \frac{1}{2}\}$$

Step3 Optimally Solve the left-guarding problem for the pair (N_L, G_L) and the right-guarding problem for the pair (N_R, G_R)

Step4 Final Solution is the combination of these two.

Result

We get a 2-factor approximation for one sided guarding problem.

Reason: Here, each point of the terrain $\mathcal{T}$ is either guarded by a member in G_L or a member of G_R or both.

4-factor approximation for General Terrain

Observation

- Let G^* be an optimum set of guards for a given instance $\mathcal{T}$, and G_L^* is the optimum solution that uses only left guards to covers the entire N_L . Then

$$w(G_L^*) \leq 2 \sum_{g \in G_L} x_g^*.$$

Reason: Put $x_{g,L} = 2x_g^*$ for guarding N_L . It is a feasible solution for guarding N_L , and its cost is $2 \sum_{g \in G_L} x_g^*$

4-factor approximation for General Terrain (contd.)

Similarly,

- If G^* be an optimum set of guards for a given instance $\mathcal{T}$, and G_R^* is the optimum solution that uses only left guards to covers the entire N_R . Then

$$w(G_R^*) \leq 2 \sum_{g \in G_R} x_g^*.$$

- A feasible solution to the one-sided discrete version with $G_L = G_R = G$, also constitutes a feasible solution to the general problem.

Result

So if we compute the one sided versions with $G_L = G_R = G$, then that would lead to a 4-factor approximation result.

Thanks